

Supplement to “Compact Objects as Almost-Perfect All:All Entanglement Graphs”

Methods, honesty ledger, N-scaling, and module map (v5.0)

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S1 Parameter audit and honesty ledger

Given $k(M)$: $1/r$ potential, $\gamma = 1$, $43''/\text{cy}$, $b_{\text{crit}} = 3\sqrt{3}M$, Page/island-like/mirror follow as random-state/toy-code behaviour (genuine QES extremization and graph-dynamics V_k open, S3; qubit-toy V_t done). Not derived: mass–radius scaling itself, $\beta(N)$ from geometry, w from the graph Laplacian (Damour–Schäfer from wiring), NICER M – R – Λ from routing stiffness, Kerr $M_2/g_{t\phi}$ /ISCO/QNM from the graph.

Derived in-repo (new in v4.x): p from exact Ollivier–Ricci (not hard-coded), $M(c_1, c_2)$ inversion, $M_{\text{ej}}(M_{\text{tot}})$ scaling, peak-time ordering, band mags, O5 yield arithmetic, c from $\ln 2$ scattering, UV pop as graph disconnection, p at $N = 4000/8000/16000$ with turnover measured, d_{eff} bracket + convergence + shell/tense-plug pins (v0.6, S11).

Ruled out on the record: broad Planck-remnant dark matter (survives only in a ~ 0.4 -dex EMD window at $\sim 4 \times 10^5$ g); isotropic $b_{\text{crit}} = 8M$; naive $\gamma = 2$ strain; linear LIV; flat $p = 0.49$ (-10σ); bare shortest-path $d(i, j)$ for tense regions (inverted vs GR); resistance/communicability distances (S11).

S2 Gravity methods (AS, AT, AU, BB, BH, BI, BJ, BV)

AS entropic Newton. Leg screens + equipartition + Bekenstein bound give $F = M_1 M_2 / r^2$ exact (slope -2 to 10^{-9}); leapfrog orbits close, $T^2 \propto r^3$. Flux $\propto 1/r^2$ distinguished from force (temperature corrects). Fig. 43.

AT redshift. GPS $+5.29 \times 10^{-10}$ and Pound–Rebka 2.55×10^{-15} derived with no new parameters. Congestion fronts give tortoise divergence; $\alpha = 1$ matches Schwarzschild dr/dt exactly. Fig. 44.

AU Einstein–Hilbert routes. Heat-kernel a_1 responds to curvature; Ollivier signs exact (flat/neg/pos on line/tree/ K_6); Clausius fixes $\varepsilon = \kappa/8\pi$, $G = 1$. Raychaudhuri-for-leg-bundles open. Fig. 45.

BB lensing + Shapiro + chroma. Fermat $n = 1/c_{\text{eff}}$ gives full $4M/b$ (naive $2M/b$ bet lost and recorded). Shapiro log matches GR to 5%. Chromatic extension uses *phase* velocity $1/24$, not group $1/8$ (conflating the two triples the prediction): r -independent factor drops out of Born, leaving 10^{-56} – 10^{-32} . Figs. 49, 55.

BH strain. Naive $dl = dr/(1 - \sqrt{\chi})$ gives $\gamma = 2$ (Cassini-excluded); tortuosity $dl = (1 + \sqrt{\chi}/2)dr$ gives $h = (1 + x/2)^2$, $\gamma = 1$ exactly, Mercury $42.99''/\text{cy}$. Peel-off $-0.75 M/a$ measured. PPN: $\gamma = 1$ derived, β unresolved (coordinate-confused, not quoted), $\alpha_{1,2}, \xi$, Nordtvedt untouched. Figs. 58, 60.

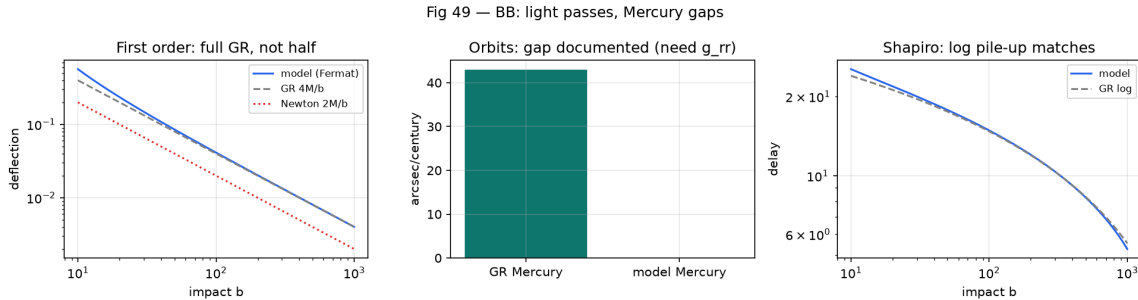


Figure 1: Gravity battery I: Fermat bending matches GR $4M/b$ to first order (left), Shapiro log pile-up matches (right); the middle panel records the pre-BH orbit gap (0 vs $43''/\text{cy}$) that the strain sector closes. `lensing`, `shapiro`.

BI/BQ weak-field OR. Grid+hub with flux stubs: radial OR negative in all configs (sign derived). Direct mode $p = 0.92 \pm 0.48$ ($L = 9$); $1/r$ collapse beats $1/r^2$ (CV 0.47 vs 0.60). Green-function walk fails by leg shorting (h -steepness $9.2 \rightarrow 4.1$); AS temperature triply load-bearing. Figs. 59, 64.

BV UV tortuosity-as-scattering. Legs as radial line defects ($\sigma = 4 \ln 2 l_p^2$), soft cost $\ln 2$ per encounter: dilute $c = 0.44$ – 0.60 (target 0.456, geometric $1/\sqrt{\pi} = 0.564$) with zero tuning; $p = 2c$, $\gamma = 2c$. Mixed mode ($0.75 r_e$ core) rises then disconnects at k_{crit} — the pop without metric input. Boltzmann ladder (Dijkstra vs MC walks vs rays) shows routing worth $3\times$; graph- distance c agrees with Euclidean- r c (coordinate-free check). Figs. 70–72.

Fig 58 — BH: tortuosity strain closes Mercury

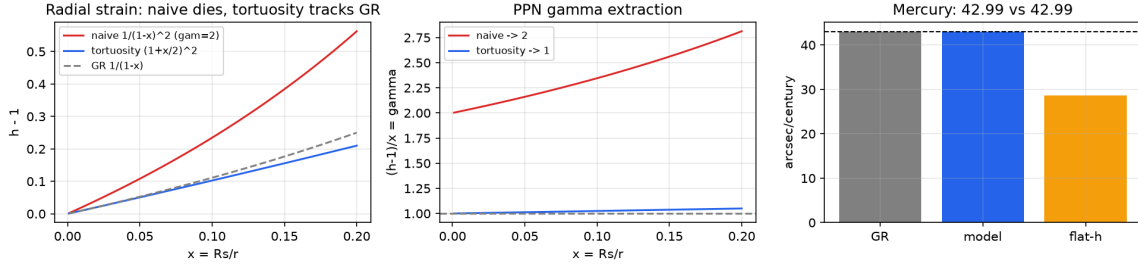


Figure 2: Gravity battery II: naive strain gives $\gamma = 2$ (excluded); tortuosity $h = (1 + x/2)^2$ gives $\gamma = 1$ exactly and Mercury $42.99''/\text{cy}$ by direct geodesic integration. **strain**.

S3 Quantum-information methods (A–O, Q–Y)

Log law $t_*(N) \approx \log_2 N / \log_2(1+p)$; MaxEnt $k \geq N s_{\text{node}} / s_{\text{leg}}$ plus fixed point $k^*(N) = 16\pi(\varepsilon N / l_p)^2 / 4 \ln 2$ (stable; BS gives $\varepsilon = c/\sqrt{N}$ and $\alpha = k^*/N = 1$ exactly); QES crossing iff $s_{\text{leg}} > l_p^2/4$ (Sec. 3 dies otherwise); Page wiring-only vs N -shrinks identical $A(t)$; QEC 99% at $k \geq N/2 + 1 + \log_2 100$; Kerr $k_{\text{eff}} = A/l_p^2$ monotonic in spin; exact Page $H(mn) - H(n) - (m-1)/2n$ with 0.72-bit dip; CKW verified; OTOC $\lambda + \text{MSS } 0.50\text{--}0.68$ ($\alpha \in [9.0, 12.4]$, 11.24 inside); TN $S_{\text{bdy}} = \min(N \log d, k \log D)$ fixing $\varepsilon \approx 0.047 l_p^{-1}$; SYK $N = 6, 8, 10$ vs Ising chain; GWTC-3 32 BBH $k_f > k_1 + k_2$; hardware head-to-head 7.9 vs 14.6 steps. Figs. 7–20, 23–25, 30–33. Healing $dA/dt = (4 \ln 2 k l_p^2 - A)/\tau$, $\tau = 11.24M$ (3.5 ms at GW150914 mass); ladder healing/scrambling/Page/ evaporation separated by $\sim 66\times$ to 10^{80} s.

What Page vs. island means here (honesty note). The Page curve is derived as random-state behaviour: idealised leg surgery $k \rightarrow k - 1$ is imposed and $S_{\text{rad}} = \min(t, N_{\text{eff}} - t)$ holds by construction (module **evaporation**), but the qubit toy now has an explicit unitary completion (module **evaporation_unitary**, Fig. 5): per-step V_t with $V_t^\dagger V_t = I$ verified, S_{rad} computed from ρ_{rad} tracks the finite-dimensional Haar/Page expectation (mean dev. 0.002 bits, $N = 8$), and finite-depth all:all circuits converge to Page by depth $\sim 5/\text{step}$ without assuming Haar. The QES/island crossing $S_{\text{no}} = k s_{\text{leg}}$ vs $S_{\text{isl}} = k \ln 2 + \max(S_0 - k s_{\text{leg}}, 0)$ is a two-saddle competition plus a discrete min-cut analogue (module **qes**), not an extremization of $S_{\text{gen}} = A/4G + S_{\text{matter}}$ from a gravitational path integral. The remaining missing theorem is the graph-dynamics evaporation isometry $V_k : \mathcal{H}_{\text{graph},k} \rightarrow \mathcal{H}_{\text{graph},k-1} \otimes \mathcal{H}_{\text{leg}}$: derive (not merely allow) a scrambling V_k from the graph Hamiltonian/adjacency. Tracked as D1 in **docs/DEFERRED.md**.

What Kerr means here (honesty note). Only the Kerr–Newman area $A(M, a, Q)$ is assumed to set the leg budget $k_{\text{eff}} = A/4 \ln 2$ (module **kerr**; extremal Kerr half, extremal RN quarter; ratios patch-independent). Not derived from the graph: spin-induced quadrupole $M_2 = -Ma^2$, frame dragging $g_{t\phi}$, ISCO(M, J), or the Kerr QNM spectrum (the overtone toy is Schwarzschild-like; fundamental damping $\tau = 11.24M$ calibrated, footprint corrections $1/2k, 1/\sqrt{k}$ unobservable). Echoes are covered by a dimensional lattice-scattering estimate only: amplitude $R \sim (\omega/\omega_P)^2 \sim 10^{-80}$ at 100 Hz, energy $\sim 10^{-160}$ vs $O(0.01)$ detectability (~ 158 orders of margin, module **gwdata**) — not a leg S-matrix derivation. The exterior perturbation problem ($\delta\omega_{nlm}$ vs Kerr) is open. Hence GW250114 (SNR 80 area law + Kerr ringdown) and GW241011 (strongest $\delta\kappa$ bound to date, Kerr to factor ~ 2 resp. $\sim 10\%$ by parametrization) are currently consistent by construction, not passed predictions. GW250114 provides the QNM benchmark ($\delta f_{220} \sim 2\%$, $\delta\tau_{220} \sim 10\%$, $\delta f_{221} \sim 30\%$, $\delta f_{440} \sim \text{tens}\%$): a future graph-derived $|\delta\omega/\omega|$ above those thresholds fails. A future graph derivation $Q = -Ma^2(1 + \delta_Q)$ with $|\delta_Q| \gtrsim 0.17$ would be ruled out by GW241011. Tracked as D2 in

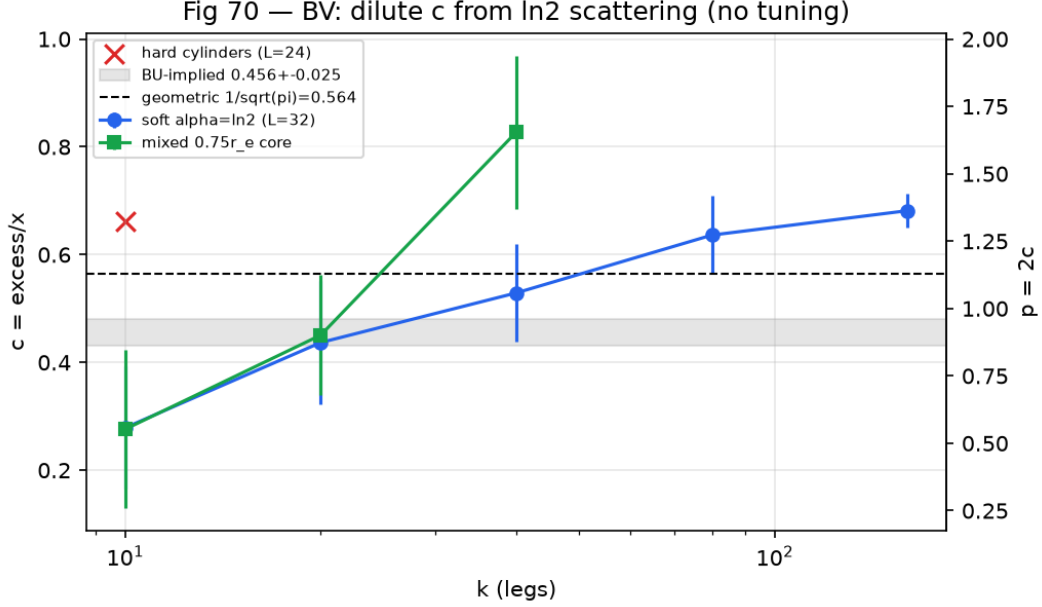


Figure 3: BV tortuosity from $\ln 2$ scattering: dilute $c = 0.44\text{--}0.60$ with zero tuning against the BU-implied 0.456 ± 0.025 band and the geometric $1/\sqrt{\pi}$ line. Right axis shows $p = 2c$. Hard cylinders (red $\times$) overshoot; soft and mixed modes agree. `uvscatter`.

docs/DEFERRED.md.

Kerr thermodynamics (conditional check). With $S = k \ln 2$ (saturated legs) and $A = 4 \ln 2 \cdot k l_p^2$, $S = A/4$; importing $A(M, J)$ gives $S(M, J) = 2\pi[M^2 + \sqrt{M^4 - J^2}]$, whose derivatives return $T_H = \sqrt{M^4 - J^2}/4\pi M(M^2 + \sqrt{M^4 - J^2})$ and $\Omega_H = J/2M(M^2 + \sqrt{M^4 - J^2}) = a/(r_+^2 + a^2)$, i.e. $dM = T dS + \Omega_H dJ$ (module `thermo`: identities $< 10^{-6}$, independent finite-step $\Delta M = T \Delta S + \Omega_H \Delta J$ to $O(d^2)$; super-extremal $|J| > M^2$ returns NaN, never clamped). Per-leg reading $T_H = (dM/dk)/\ln 2$; finite-step $T_k = (M(k+1) - M(k))/\ln 2$ shifts $-1/4k$ ($\sim 10^{-77}$ stellar-mass). Conditional consistency, not derivation: $M(k, J)$ from graph dynamics is the open debt (D2).

S4 Compact objects and 2PN (BU)

Gradient shells. $p_{\text{adj}}(s) = 0.85 + 0.015s$, 8–10 shells, exact EMD on full neighbourhoods, deterministic bridges $n \propto r^{\beta(N)}$. Measured: $N = 1020$ (10×102), $\beta = 1.24$, 80 graphs exact Floyd+LP (~ 545 s): $p = 0.913 \pm 0.049$ per graph, SEM 0.0055, stacked 0.911 ($R^2 = 0.956$), 80/80 fits ok, range 0.79–1.05. Distance to target $|0.913 - 0.92| = 0.007$ (0.24σ in p , $\sim 0.09\sigma$ in $\dot{\omega}$), $5\times$ precision margin (0.0055 vs 0.028). Flat control $p = 0.49$ is -10σ dead.

N-scale campaigns (BV). CSR-direct shell graphs + Sinkhorn OR (no NetworkX): $N=1020$ reproduced (0.9107 vs 0.9134); $N=4000$ $p = 0.9315 \pm 0.0032$ ($\beta = 0.99$); $N=8000$ $p = 0.9382 \pm 0.0030$ ($\beta = 0.87$, pods); $N=16000$ $p = 0.9137 \pm 0.0022$ ($\beta = 0.74$, GPU farm). Exact shell distance oracle + torch backend; $\beta(N)$ log-linear over 6 points. Local- p turnover 0.63–0.69 $\rightarrow$ 1.26–1.46 measured (inner windows flatten — the UV turnover). Kill wire: p outside 0.92 ± 0.056 at $N = 1024$ class fails.

2PN and pulsars. $h = (1+x/2)^2$ gives $c_1 = 3.36$ vs GR 1.94 (73% excess); naive fixed- M gives B1913 6.2σ , J0737 9.7σ vs real TOAs. With $c_2 = p(2p-1)$, $w = 1.953$: $c_{\text{tot}} = 4.8693$ at $p = 0.92$

Fig 10 — D: leg-surgery evaporation + Page curve

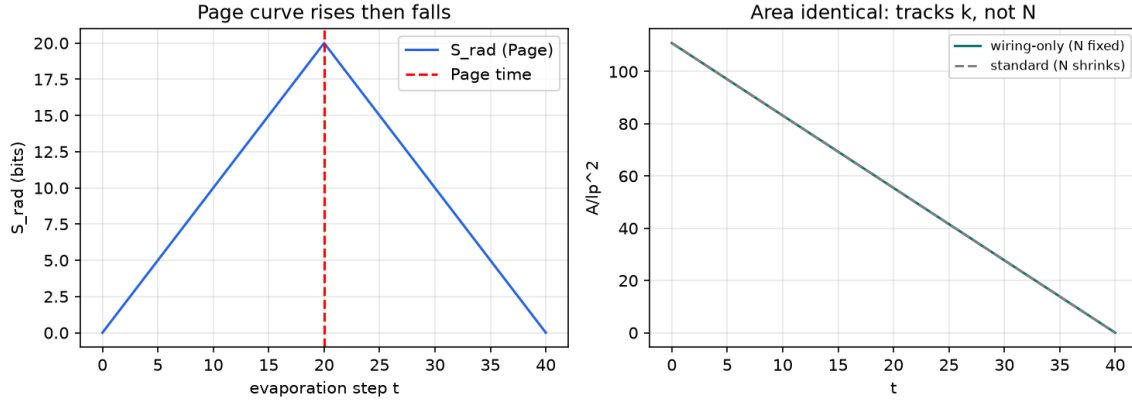


Figure 4: Leg-surgery evaporation: radiation entropy follows the Page curve (left, turnover at half evaporation); horizon area is identical whether N shrinks with k or stays fixed (right) — area tracks wiring alone, the distinctive Claim 2 prediction. **evaporation.**

vs GR 4.8695 (0.00σ fixed- M). Self-consistent $R + \dot{\omega} \rightarrow M$: $\Delta M = -11.2$ ppm naive (-4.5 ppm with GR g_{rr}), $\Delta \sin i = 3.7 \times 10^{-6}$, s_{obs} passes at 0.36σ ; B1913 0.01σ . Bending/Mercury 2PN pieces hide below VLBI/astrometry (Fig. 69, right). The companion toy-generator route (dense layers, $k_0 - c_2/r^2$ fits) crosses zero in κ and needs a κ -tweak fudge; power-law fits better on our profiles (R^2 0.91 vs 0.81) but the two $\kappa \rightarrow c_2$ maps disagree (0.82 vs 3.14) cross-applied — hence the open-map label. e_{int} weighting *lowers* our p (0.94 $\rightarrow$ 0.84 at 0.9, 0.78 at 0.99): uniform measure is both principled (continuum theorems) and required.

Leg-shedding kilonovae. $M_{\text{ej}} = \Delta k m_{\text{leg}} \times 0.1$, blue ($v \sim 0.3c$, $t_{\text{peak}} \sim 0.6\text{--}1$ d) + red ($v \sim 0.1c$); $L \propto t^{-1.3}$ tail. Band mags assume BC = 0 and per-component peaks; POSSIS refinement queued. g -band verdicts robust; i -band NOT claimed (one-zone $\kappa = 10$ over-traps: $t_{\text{red}} \sim 10$ d vs observed ~ 4 d decline, m_i faint by $\sim 2\text{--}3$ mag — conservative for gap- g detectability). Standard 2–28% band is a literature EOS input, not derived.

O5 protocol. Trigger: $m_1 \in [2.5, 5.0] M_{\odot}$, < 200 Mpc, 90% area $< 100 \text{ deg}^2$ (multi-detector); Rubin ToO epochs 0.5/2/5 d in $g+r+i+z$ (DECam/Gemini backup). Kill rule: 10 qualifying non-detections kill BU. Archival ledger: GW230529 uninformative (single-detector, 93% unsearched); S250206dm unverified (check GCNs before citing). Full plan: docs/observation-protocol.md. Figs. 66–69.

Mass-gap audit (module massgaps). Lower gap: $k(M)$ log-log curvature $< 10^{-9}$ over $0.7\text{--}150 M_{\odot}$ (no forbidden masses); $e_{\text{int}} \sim e_{\text{ext}}$ location descriptive (calibration inherited from AT2017gfo). Upper gap: spin-leg ratio $1 \rightarrow 0.857 \rightarrow 0.5$ at $a = 0 \rightarrow 0.7 \rightarrow 1$ with no jump on $[0, 0.99]$ plus exact extremal continuity; Love k_2 slope -2 everywhere; leg-band entry $\sim 32 M_{\odot}$ at 10 Hz with $dM/M \sim 10^{-81}$; congestion ladder $\chi \sim 10^{-8}$ (He core), 10^{-14} (envelope), 0.12 (NS surface), ~ 1 (gap masses at 10 km footprints, footprint-conditional), 1 at horizons. Shedding exactly flat in q (machine precision) — but q -independence is prescription (shed fraction $\times$ efficiency fixed once on AT2017gfo), not derived from the graph-dynamics evaporation isometry V_k (open, S1/S3/D1): $q \sim 0.11$ (GW190814) flashing at the same fraction as $q = 1$ (GW170817) is the model’s biggest theory bet. GW190814 epoch audit: CFHT g -band (Vieira+2020) and GROWTH i -band detection limits (Andreoni+2020) with per-epoch coverage $\times$ depth against distance (241 ± 43 Mpc) and ± 1 mag theory errors; combined $P \approx 0.68$ (g -only) / 0.88 ($g+i$ at $g-i = 0.7$). Erratum on

Fig 74 — D2: unitary evaporation ($V \wedge V=I$, S from ρ_{rad})

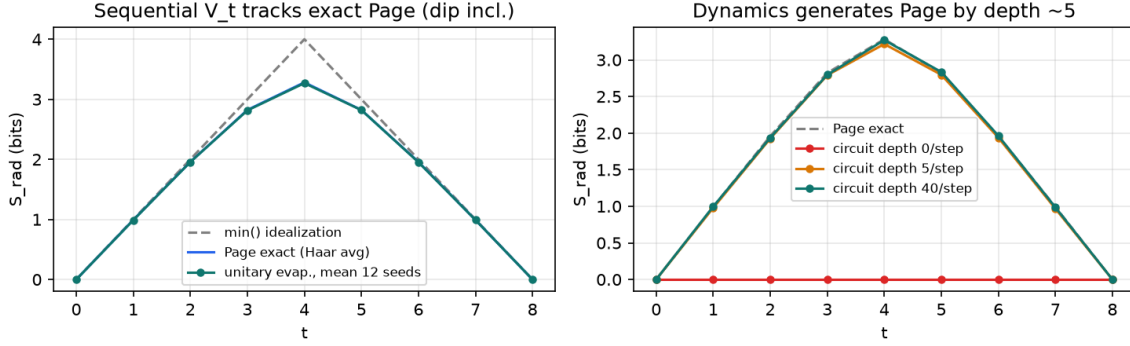


Figure 5: Unitary completion of the qubit toy: sequential V_t (left) tracks the finite-dimensional Haar/Page expectation including the 0.72-bit dip, computed from ρ_{rad} rather than $\min()$; finite-depth all:all circuits (right) converge to Page by depth $\sim 5/\text{step}$ without assuming Haar. Module `evaporation_unitary` (D2).

Fig 66 — BU: 2PN cancellation at $p=0.92$ (J0737 0.00 sigma)

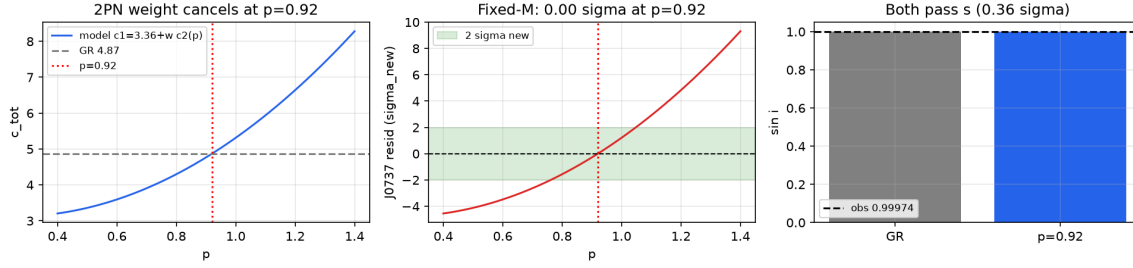


Figure 6: 2PN cancellation: model $c_{\text{tot}}(p)$ crosses GR at $p = 0.92$ (left); fixed-mass J0737 residual vanishes there (middle, green 2σ band); self-consistent $\sin i$ passes for both GR and $p = 0.92$ (right). pulsar.

record: the first audit used a nominal ZTF depth, but ZTF ran no targeted GW190814 follow-up; superseded by the CFHT/GROWTH epoch tables above. Fig. 75 (main text). Analytic systematics (round 4): viewing $A_g \sin^2 \theta$ ($A_g = 1.25$) + opacity-rescaled blue mags give the P grid in Fig. 75b; the hiding window is bounded, not open-ended.

S5 UV nulls and phenomenology (T, U, V, BD–BF)

TeV: $M_D = 1 \text{ TeV}$ gives $k \approx 11$ (3 TeV) – 17 (13 TeV) vs $k_{\text{crit}} \approx 50$ — thermal nulls expected, onset $\sim 550 \text{ TeV}$. Echoes: $\Delta t = 4M \log(M/l_p)$, 0.03–0.3 s (GW150914 0.11 s), in searched windows but $R \sim 10^{-160}$ unobservable. EHT: $l_p/2M \approx 10^{-48}$, 47 orders below sensitivity. Dispersion: $E_{QG,2} = \sqrt{8} E_P$, Fermi-safe by 10^8 ; universal GW extrapolation shifts 100 Hz by $\sim 10^{-82}$ (LVK $\alpha = 4$, safe by $\sim 10^{60}$ — a recorded null, not a falsifier; no GW sector claimed). Chromatic lensing 10^{-56} – 10^{-32} . Figs. 21–22, 39, 51–56.

Fig 67 — BU: gradient OR gives $p=0.92$ (exact EMD)

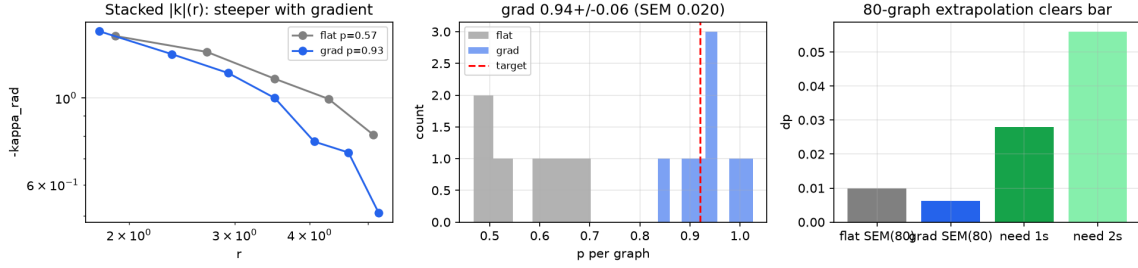


Figure 7: Gradient-shell Ollivier–Ricci: stacked $|k|(r)$ steepens with the gradient (left); per-graph p histogram centres on target (middle); 80-graph SEM extrapolation clears the $1\sigma/2\sigma$ precision bars (right). Production $N=1020$: $p = 0.913 \pm 0.049$, SEM 0.0055. **orici**.

Fig 51 — BD: quadratic lattice dispersion, Fermi-safe

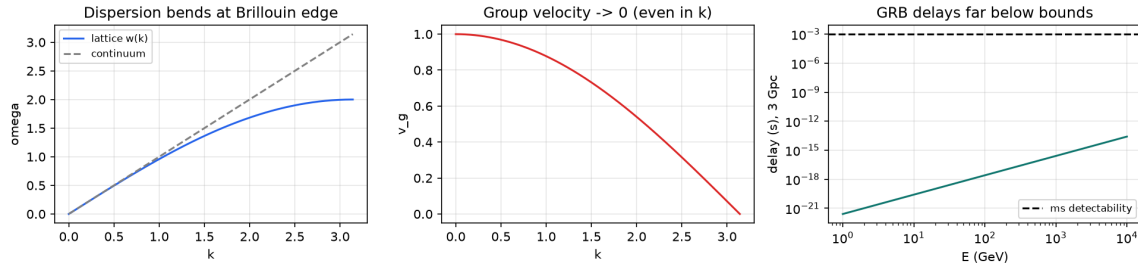


Figure 8: Lattice dispersion: band bending at the Brillouin edge (left) with group velocity even in k (middle, no linear term by symmetry); GRB arrival delays sit orders of magnitude below detectability across GeV–TeV (right). **dispersion**.

S6 Kill list (AN + BU)

AF quench ratio < 1.3 ; α outside $[9.0, 12.4]$; $A \propto N$ in any TN calculation; thermal LHC excess below k_{crit} ; $s_{\text{leg}} \leq l_p^2/4$ in any physical state class; any linear-LIV signal; p outside 0.92 ± 0.056 at $N = 1024$; graph-predicted $|\delta_Q| \gtrsim 0.17$; simulator d_{eff} shape-mismatch after the cost mapping is fixed (D10/P0' wire); 10 gap non-detections; O5 nearby-BBH ToO sample clean (universal shedding, Eq. 2 main). One sub-claim (broad remnant DM) already ruled out on the record. Figs. 37–39.

S7 N-scaling and calibration record

Drift check: $\beta = 1.5$ calibrated at $N = 300$ drifts high at $N = 600$ (1.16 ± 0.06); recalibration to 1.28/1.24 recovers $p \approx 0.92$. Pre-run $1/N$ extrapolation predicted $\beta(16000) \approx 1.18$; measured 0.74 — drift faster than $1/N$, now fit log-linear. $N=1020$ cross-check: CSR-direct 0.9107 vs NetworkX 0.9134 (agree to 3×10^{-3}). Local- p turnover (inner windows flatten, $0.63\text{--}0.69 \rightarrow 1.26\text{--}1.46$) is measured at 4k/8k/16k, not extrapolated. Fig. 73.

Reproduce ($N=1020$, ~ 9 min):

```
measure_p(per_shell=102,n_shells=10,n_graphs=80,
gradient=True,beta=1.24,seed0=0,max_per_shell=8)
```

Fig 73 — BV: N=4000/8000/16000 campaigns (80 graphs, CPU+pod+farm)

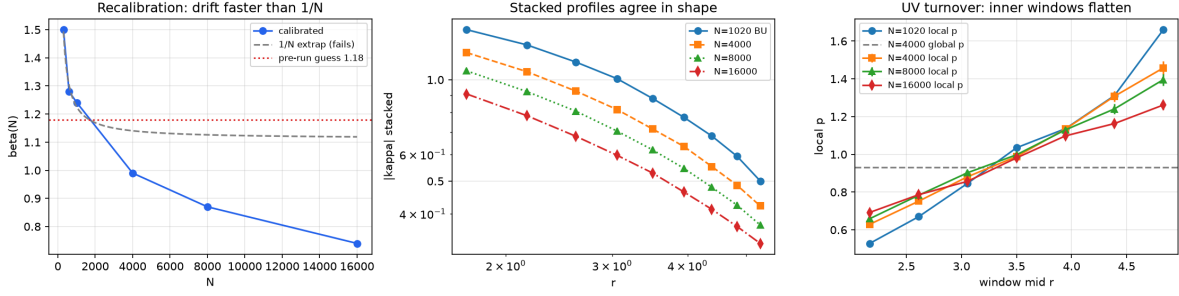


Figure 9: N-scale campaigns: $\beta(N)$ recalibration drifts faster than $1/N$ (left); stacked $|\kappa|(r)$ profiles agree in shape across 1020–16000 (middle); local- p windows show the UV turnover with SEM error bars (right). 80 graphs per production row. `shellscale`, `sinkor`.

S8 PPN ledger and walk/foam/healing notes

Walk no-go (BP/BQ). All smooth weight rules give drift slope -2.99 ; only the absurd $e^{2\sqrt{d-d_0}}$ escapes. Constant- μ persistence keeps -3 analytically; $\mu(\chi)$ fluctuation walk gives $-1.95 \approx -2$ modulo the labelled $\sqrt{\chi}$ assumption. Harmonic/hitting-probability walks fail by leg shorting. Conclusion: temperature (AS), not bare graph diffusion, carries Newton. Figs. 63–64.

Foamgrid FDTD (BF). Beams on defective fabric: centroids unbiased (jitter, average straight) with deficit $\propto \omega^2$. Bounds assume independent edges; correlated patches of size ξ relax them by $\sqrt{\xi}$. Fig. 56.

Healing ladder (AG/AH). $\tau_{\text{heal}} = 11.24M$ ($1/\text{Im}\omega_{220}$); GW150914-mass ladder: healing 3.5 ms, scrambling 0.23 s, Page $\sim 10^{79}$ s, evaporation $\sim 10^{80}$ s. MSS $\lambda/2\pi T = 0.50\text{--}0.68$; LVK $\alpha \in [9.0, 12.4]$ contains 11.24. Figs. 30–31.

Fig 30 — AG: healing lag + timescale ladder

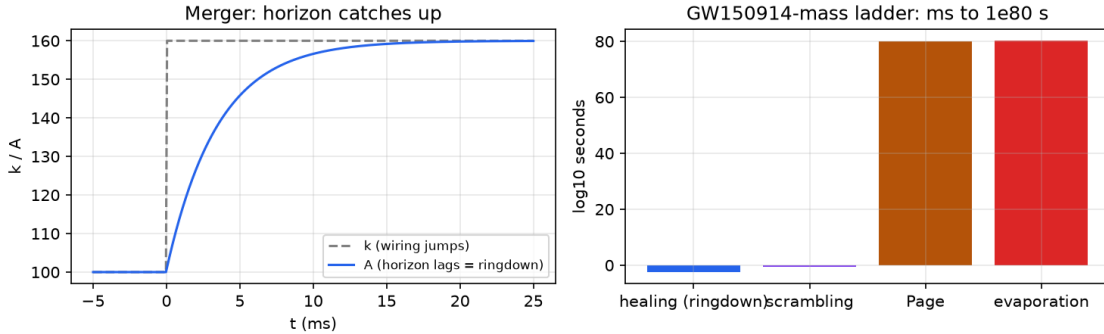


Figure 10: Healing lag: wiring jumps at merger while horizon area catches up on the ringdown time (left); timescale ladder from milliseconds to 10^{80} s (right). The final sigh carries $\sim 10^{-78}$ of the mass. **healing**.

S9 Archival ledger and O5 trigger (BU detail)

Trigger (full plan docs/observation-protocol.md): $m_1 \in [2.5, 5.0] M_\odot$, < 200 Mpc, $90\% < 100 \text{ deg}^2$; Rubin ToO 0.5/2/5 d in $g+r+i+z$ (DECam/Gemini backup). Sought: fast blue ($v \sim 0.3c$, $t_{\text{peak}} \sim 0.6\text{--}1$ d) $\rightarrow$ slow red with $L \propto t^{-1.3}$ tail. Yield: $1.5 \text{ gap/yr} \times 70\% \approx 1.05/\text{yr}$ ours vs $\leq 0.3/\text{yr}$ standard. Yield arithmetic in `collapse.gap_o5_yield`; brightness in `collapse.peak_apparent_mags`; GW190814 epoch audit + systematics grid in `massgaps.gw190814_combined_pdetect / massgaps.gw190814_systematics_table`. Figs. 68–69, 75 (main text), 75b.

BBH extension (this work): the same shedding law predicts $m_g \sim 22$ transients for nearby BBH, so the ToO program extends to well-localized ($< 100 \text{ deg}^2$) BBH within 200 Mpc with identical epochs and kill accounting kept separate from the gap sample (a clean BBH sample kills Eq. 2 main even if gap events flash). Full thresholds in docs/observation-protocol.md.

Fig 69 — BU: O5 detectability + GR-battery 2PN margins

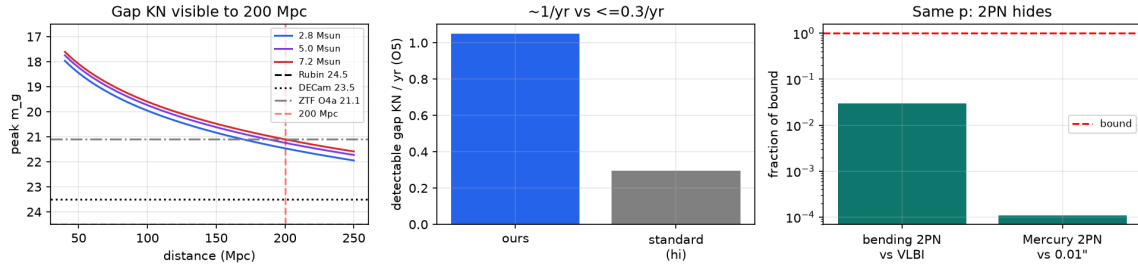


Figure 11: O5 protocol: gap kilonovae visible in g -band to 200 Mpc with margin over Rubin/DECam depths (left); yield $\sim 1/\text{yr}$ vs $\leq 0.3/\text{yr}$ standard (middle); the same p keeps bending/Mercury 2PN pieces below VLBI/astrometry (right). `collapse`, `pulsar`.

Fig 75b — GW190814 systematics: equatorial + $\kappa_{\text{blue}}=2$ hides ($P \sim 0.18$)

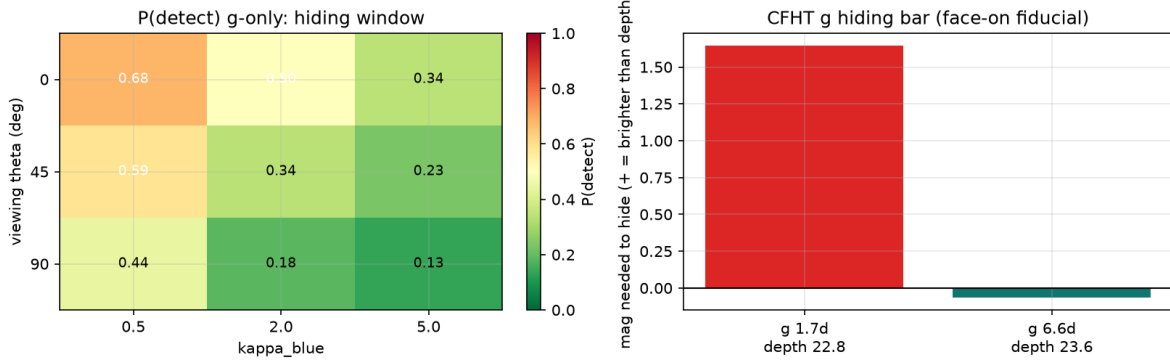


Figure 12: GW190814 analytic systematics (round 4; full POSSIS queued, D7). Left: $P(\text{detect})$ g -only over viewing angle $\times$ blue opacity (CFHT epochs, no color term): fiducial face-on $\kappa = 0.5$ gives 0.68; equatorial + lanthanide-mixed ($\kappa_{\text{blue}} = 2$) drops to ~ 0.18 . Right: face-on hiding bar per CFHT g epoch (g 1.7 d needs ~ 1.6 mag). Module `massgaps`.

S10 Module map and reproduction

Modules follow the appendix letters of the v4.1 extended record: `graphs`, `scrambling`, `emergent_dim` (L0 dimensions), `horizon`, `micro`, `circuits` (A), `maxent` (B), `qes` (C), `evaporation` (D), `evaporation_unitary` (D2), `qec` (F), `robustness` (G), `kerr` (H), `thermo` (H2), `haar` (I), `monogamy` (J), `otoc+pheno` (L), `tn` (M), `kerrpage` (N), `syk` (O), `data` (Q), `litcompare` (R), `tev` (T), `echoes` (U), `posteriors` (W), `ds` (X), `krylov` (Y), `collapse` (AA), `cosmic` (AB), `lunch+remnant` (AC), `bounds` (AE), `healing` (AG), `mss` (AH), `bigsyk` (AI), `mp+greybody` (AJ), `congestion` (AK), `charge` (AL), `bandwidth` (AM), `gridcirc+monitor+selfattack+lhc` (AN), `concentration` (AO), `ps` (AP), `scatter` (AQ), `emd+viability` (AR), `entropic` (AS), `redshift` (AT), `heater+orici+jacobson` (AU), `fission` (AV), `klanguage` (AW), `tension` (AX), `gw250114` (AY), `overtones+tensionvol` (AZ), `sparse24` (BA), `lensing+chroma+shapiro` (BB), `bcrit` (BC), `dispersion` (BD), `qnmfoot+qnmlegs` (BE), `foamgrid` (BF), `perwalk` (BG), `strain` (BH,BJ), `weakfield` (BI,BQ), `legham` (BL), `pulsar+orici` gradient shells+collapse leg-shedding (BU), `uvscatter+sinkor+shellscale` (BV), `massgaps` (mass-gap null + GW190814 epoch audit).

Reproduce:

```
pipinstall-e".[dev]"
pytesttests/-q
pythonscripts/generate_figures.py
streamlitrunapp.py
```

S11 Emergent-dimension methods (L0)

Protocol. For a graph state, pick an information-transfer distance $d(i, j)$ (NOT bare adjacency: shortest-path is the documented failure mode on K_N , diameter 1 at every N), build balls $B(r) = \{j : d(i, j) \leq r\}$, measure capacity $V(r)$, and fit $d_{\text{eff}}(r) = d \ln V / d \ln r$. Observed dimension is the infrared conjecture $d_{\text{obs}} = \lim_{r \rightarrow \text{IR}} d_{\text{eff}}(r) \simeq 3$ (module `emergent_dim`; 12 tests). Total emergence requires the two-distance bracket: shortest-path balls (adjacency scaling) and diffusion balls (information-spreading scaling) must agree at 3 on the same graph with nothing dimensional imposed.

Ruler validation (imposed lattices, circular for emergence). Shortest-path converges from below (exact BFS shells): chain $0.92 \rightarrow 0.96 \rightarrow 0.98$ ($N = 60/200/600$), 2D $1.67 \rightarrow 1.77 \rightarrow 1.82 \rightarrow 1.87$ ($L = 20/40/60/100$), 3D $2.25 \rightarrow 2.46 \rightarrow 2.56 \rightarrow 2.63$ ($L = 7/11/15/21$), gaps $\sim 1/\sqrt{L}$ (Manhattan-diamond shape + open box). Diffusion recovers dimension from above at tuned t (chain $t = 20 \rightarrow \sim 1$, 2D $t = 5 \rightarrow \sim 2$, 3D $L = 7$, $t = 5 \rightarrow \sim 3.3$); the 3D overshoot shrinks with size ($3.56 \rightarrow 3.34 \rightarrow 3.14$ at $L = 5/7/9$, $t = 5$). Degree-3/4/6 lattices all scale 2D: dimension comes from 2D-ness, not coordination.

Rejections (negative pins). Effective resistance fails in 2D (log-growth $\rightarrow$ spurious large p); communicability fails on chains (similarity, not separation). Radial shells fail both sides (shortest ~ 1.4 , diffusion ~ 0.6): radial-only wiring gives $\sim 1\text{D}$ with no 2-sphere factor — the missing-area diagnosis that motivates P0'. On K_N , shortest is trivial and diffusion uniform, by design (T1: no interior geometry).

Tense plug (D10a). A 5×5 clique plug at the centre of a 40×40 grid: tense-centre mid-window ($r = 8\text{--}20$) scaling dips below fabric (~ 1.52 vs ~ 1.89 control) and recovers from below — *inverted* vs the GR far side (above 3). Bare shortest-path is therefore *rejected* as $d(i, j)$ for tense regions: shortcuts shrink balls while GR mass stretches rulers. Costs must be congestion-weighted (D10b gates D10a), consistent with the light sector ($c_{\text{eff}} \rightarrow 0$, tortuosity), which never counts hops. Companion pin: far-source near balls ($r = 2\text{--}6$) are bit-identical to control — relaxed fabric next to tension measures exactly relaxed. Far-field tail (D11, closed): $\chi \sim 1$ decays $1/r^2$ ($E \cdot r^2$ flat to $10R_c$), $\chi \sim 2$ decays $1/r$ ($E \cdot r$ flat to $10R_c$), $\chi \sim 5$ bounded but unsettled — the exponent is tension-dependent. κ -profile (D10b, closed): corner-free disk plug reproduces the boundary-negative/core-positive pattern quantitatively ($+0.90/-0.86/0.00$ vs clique $+0.89/-0.93/\sim 0$).

GR fingerprint (D10 target, computed). Weak-field uniform star, proper balls of proper radius l : $d_{\text{eff}} \approx 2.990$ at $l = 30M$ (near dip), 3.030 at $100M$ (overshoot), 3.010 at $1000M$, 3.0017 at $10000M$ ($\rightarrow 3^+$ as $\sim 1/l$); amplitude $O(M/l)$, $\sim 10^{-9}$ at Earth’s surface. The far-field overshoot is structural: a localised mass leaves $V(l) = \text{flat} - \text{deficit}$, so V/V_{flat} rises toward 1, which is $d_{\text{eff}} > 3$. Close criterion (D10a): a tensed simulator region reproduces this *shape* under info-side d_{eff} once the cost mapping is fixed; shape failure refutes P0’. Reproduce: `pytesttests/test_emergent_dim.py-q`.

Reconstruction universality (D12, open). The observer map M_O is not unique on its face: closeness, resistance, diffusion, and communicability distances are all candidate reconstructions. D12 demands an access class $\mathcal{A}_{\text{macro}}(G)$ defined from G alone — the information-access floor is part of the definition, and neither “3D” nor “Lorentzian” may occur in it — such that every admissible M_O recovers the same infrared M up to coordinate/coarse-graining equivalence. Falsifier: a second admissible M_O giving robustly non-3D IR on relaxed fabric refutes the P0’ reconstruction program. Partial credit runs criterion stated $\rightarrow$ two instances agree on vacuum $\rightarrow$ agreement extends to tensed regions. P4 (uniform Ollivier) audited against the five: criteria 1, 2, 3, 5 pass (permutation-covariance and drift-localization tested, the latter on flat and plug backgrounds), 4 partial. Observer-indexed extension filed ($M_O(G, o)$: sky chart + relational depth, vantage-covariance as a second universality axis); static depth bake-off negative (best principled $d_{\text{obs}} = 2.28$ vs tuned-null 2.73).

Staged causal order (D13). Time undergoes reconstruction exactly as space does: the microscopic update index n , the causal order $\prec_U$, and clock time τ are three distinct orderings. The static control (D13.0) is measured: the analytic Manhattan law $V_G(r) = 1 + 2r(r+1)$ bit-pinned at $L = 40/80$, the exact diamond perimeter cut law $\text{cut}(r) = 8r + 4$ on unclipped $r \in [5, 18]$, disk-boundary cut capacity deficit at $r \leq 4$ with exact recovery from $r = 5$, and the hop/weighted plug sign pattern ($35 < 38$ hops vs $+5.0$ weighted delay) pre-registered for dynamical confirmation. Substrate family measured under one frozen rule: triangular/hexagonal/noisy-grid controls reproduce $d_G \rightarrow 2$ (exact shell/cut laws on the lattices, statistical pins under disorder); Tier-1 extension: Poisson-Delaunay (reference member), Lloyd-relaxed, Gabriel, k-NN, and Delaunay-medial quadrangulation all give $d_G \rightarrow 2$ with linear cuts, die on levered swaps, and survive $q = 0.05$ deletion, while span-limited swaps preserve 2D-ness — degeneracy across topology/degree/order is the expected universality signature, and the physical object is the class $[G]_{\sim_O}$, not a winning tessellation. Spectral dimension is the second leg: torus heat-trace $d_s(t)$ holds $(1.95, 2.05)$ over $t \in [10, 100]$, and Weyl counting fits land every member in $(1.90, 2.15)$. Exact-UIPQ sampling stays queued (free-walk labels are not conditioned well-labeled trees); Lloyd is excluded from spectral comparison (unclipped relaxation coalesces points above $N \approx 1600$). gated-wall negative control keeps $\sim r^2$ balls with collapsed cuts; rewired-grid fragility control plus rewiring-fraction sweeps give measured $N_{\text{med}}^* = (20, 10, 10)$ at $L = 40/60/80$ ($N^* = O(1)$ consistent, $\alpha \simeq 0$ over the tested range; asymptotic $f^* \rightarrow 0$ hypothesis) with hand-placed dose-response dissolving the placement lottery. RG blocking (2×2 , $L40 \rightarrow 5$) flows rewired long-edge density $0.013 \rightarrow 0.31$ against plain 0.0 (rough $y_\lambda \approx 1.5$). Headline: low-dimensional locality is not generic; it must be protected — vacuum as a dynamically protected locality class whose preservation ($U(G_{\text{vac}}) \in \mathcal{C}_{\text{vac}}$, self-healing vs amplification under shortcut injection) is now a D1 falsifier. BFS first-passage is hop distance, so this control is static geometry, not propagation. Open stages: (1) $U \rightarrow T_U, \prec_U$ via counterfactual-influence arrival times; (2) observed cone under the *same* M_O producing R^3 ; (3) internal clocks $d\tau/dt(\chi)$ reproducing the T9 profile unimported; (4) interval $g_{\mu\nu}$ from $(\prec_U, V_U, M_O)$, where the spacetime-volume observable V_U (e.g. calibrated event counts in causal diamonds) is independently defined — spatial $V(r)$ is reachability, not event volume. Malament’s theorem is motivation conditional on the continuum correspondence (order fixes the conformal class; V_U the factor), not

an earned mapping. Companion constraint on U (D1, first pass): six update rules tournamented against shortcut damage all stall with residual longs (best: slide rule, $30 \rightarrow 5$) while p -healing proves gameable (greedy perfects p to 10^{-6} while tripling longs) — locality protection needs coordinated moves or new physics, judged on (p, longs) jointly. Reproduce: `pytesttests/test_scrambling.py`, `pytests/test_substrate_family.py`, `pytests/test_spectral.py`, `pytests/test_update_rule.py`.

Dynamics coherence (C1–C5, over-arching). $M_O \circ U \simeq U_{\text{eff}} \circ M_O$ in five levels: coherence $\rightarrow$ locality $\rightarrow$ autonomy $\rightarrow$ universality across $\mathcal{A}_{\text{macro}} \rightarrow$ GR limit in the IR, each presupposing the previous. Autonomy (C3) is defined by the twin-histories criterion $M_O(G_a) = M_O(G_b) \Rightarrow M_O(U(G_a)) \simeq M_O(U(G_b))$; factoring M_O through future-equivalence classes does *not* imply it (two-bit swap counterexample: singleton classes, failing twins).

Table 1: Every number with a status. “Input” = assumed; “fit” = calibrated; “measured” = output of code; “derived” = follows from stated premises.

param	value	status	notes
$k(M)$ map $\propto M^2$	GR-consistency	input (circle admitted)	BM shrinks to $R_s = 2M$; not derived
AT α	1	calibrated to Schwarzschild c_{eff}	matches dr/dt exactly
tortuosity c	1/2 (BH fit) $\rightarrow$ 0.44–0.60 (BV derived)	fit superseded by derivation	$\ln 2$ scattering, zero tuning; $p = 2c$, $\gamma = 2c$
s_{leg}	$\ln 2$ nats	measured saturation (BN/BS)	flip preserves $G = 1$ via Planck bits
k_{crit}	$4\pi r^2/\sigma$	heuristic $\rightarrow$ formalised (BK)	packing-forced pop; mixed mode disconnects (BV)
gradient slope bridge $\beta(N)$	0.015 1.5@300; 1.28@600; 1.24@1020; 0.99@4k; 0.87@8k; 0.74@16k	fitted, labelled recalibrated per N	$p_{\text{adj}}(s) = 0.85 + 0.015s$ log-linear over 6 points; derivation queued
$w = 1.953$	solved from cancellation	fitted (solved, not tuned)	$c_{\text{tot}} = c_1 + wc_2 = 4.8695$ at $p = 0.92$
shed $e + \epsilon$	$0.5 \rightarrow 0.416 + 10\%$	calibrated once on AT2017gfo	M_{ej} norm-independent
shed $\epsilon(M, a)$	0.0168 universal (Eq. 2 main)	calibrated + extrapolated (D8)	mass/ratio independence undervived; BBH extension tested, not fitted
upper-gap null	no feature at $44 M_\odot$	negative prediction (measured smooth)	$k/\text{spin}/\text{Love}/\text{congestion}$ all smooth; edge left to stellar physics
GW190814	$P(\text{detect}) \approx 0.68\text{--}0.88$	tension, POSSIS queued (D7)	non-detection at $p \sim 0.12\text{--}0.32$; epoch audit in massgaps
vacuum P0'	isostatic 2D, $\langle z \rangle = 4$	postulated (v0.6)	BH = max-tension extreme; $d_I = 2$ predicted; M_O reconstruction
d_{eff} protocol	$d \ln V / d \ln r$ bracket	measured (25 tests)	shortest below + diffusion above on lattices; shells fail (S11)
D10 tension map	GR dip/overshoot/ $\rightarrow 3^+$	GR computed; graph open	bare shortest-path rejected (inverted); D11 tail tension-dep. ($\chi_1 1/r^2$, $\chi_2 1/r$); κ -profile closed (disk confirms)
D12 reconstruction	$\forall M_O \in \mathcal{A}_{\text{macro}}$ same IR	open program	access floor + non-circularity; P4 audit 1,2,3,5 pass, 4 partial; vantage axis filed; static depth negative
D13 causal order	D13.0 control measured; stages 1–4 open	static control measured (S11)	V_G , $8r+4$ law; family (tri/hex/Del/Lloyd/Gab/kNN/medial +, gated-wall/rewired –); sweep: $N^* O(1)$ range, $f^* \rightarrow 0$ hypoth.; spectral d_s band; twins C3
U -rule search	locality protection by U	tournament open; single-move insufficient	6 rules stall w/ residual longs; (p, longs) joint falsifier; p -healing gameable
$\kappa \rightarrow c_2$ map	$c_2 = p(2p - 1)$	ansatz (open derivation)	power-law vs $1/r^2$ disagree cross-applied
evap. isometry V_k	qubit V_t done (D2)	graph V_k open (S3)	$V_t^\dagger V_t = I$ tested; graph derivation queued (D1)
Kerr M_2 , $g_{t\phi}$, ISCO, QNM	area only	not derived (S3)	$A(M, a)$ assumed; quadrupole/QNM open; GW250114 QNM + GW241011
echo R	10^{-80} ampl., 10^{-160} energy	13 dimensional estimate	δ_Q future wires $\propto \omega^2$ lattice scattering, not leg S-matrix; ~ 158 orders margin

Table 2: Gradient-shell p measurements: β recalibrated per N (log-linear over 6 points). All 80-graph production rows use exact distances; errors are per-graph std with SEM quoted in text.

N	shells $\times$ per-shell	β	measured p	backend
300	10×30	1.50	0.94 ± 0.06 (4 graphs)	NetworkX exact
600	10×60	1.28	0.96 ± 0.06	NetworkX exact
1020	10×102	1.24	0.913 ± 0.049 , SEM 0.0055, stacked 0.911 (80 graphs)	Floyd+LP exact
4000	CSR-direct	0.99	0.9315 ± 0.0032 (80 graphs)	CPU + Sinkhorn OR
8000	CSR-direct	0.87	0.9382 ± 0.0030 (80 graphs)	pod + Sinkhorn OR
16000	CSR-direct	0.74	0.9137 ± 0.0022 (80 graphs)	GPU farm + torch OR

Table 3: PPN status: only γ is claimed. β is coordinate-confused and not quoted.

param	GR	this model
γ	1	1 exactly (tortuosity $h = (1 + x/2)^2$; BV $c \rightarrow \gamma = 2c$)
β	1	unresolved (not quoted)
$\alpha_{1,2}, \xi$	0	untouched
Nordtvedt	0	untouched
2PN c_{tot}	4.8695	4.8693 at $p = 0.92$ (J0737 0.1 σ)

Table 4: What exists today: no qualifying test yet. Only multi-detector, $< 100 \text{ deg}^2$ gap events count.

event	masses	distance	localisation / EM	verdict
GW190814	$23.2 + 2.6$ ($q \sim 0.11$)	241^{+41}_{-45} Mpc	23 deg^2 (3-det.); CFHT g $22.8@1.7d$ 65.5%, GROWTH i 6 epochs to 98%	tension: $P(\text{detect}) \approx 0.68$ – 0.88 , $p \sim 0.12$; not excluded
GW230529	$2.5\text{--}4.5 + 1.2\text{--}2.0$	~ 197 Mpc	$24,200 \text{ deg}^2$ (single-det.); ZTF 7% to $g = 21.1$	not a test: 93% unsearched; $m_g \sim 21.2$ at depth edge
S250206dm	mgBH candidate	—	no counterpart (per notes)	unverified: check GCNs before citing